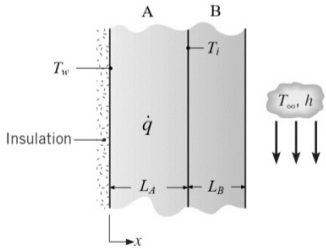
2. (40 pt) Consider one-dimensional heat conduction in a plane composite wall. Wall A has a thickness of $L_A = 30$ cm, which experiences uniform volumetric heat generation $\dot{q}$. The thermal conductivity of wall A is k_A. The left surface of wall A is thermally insulated. Wall B has a thickness of $L_B = 10$ cm and thermal conductivity $k_B = 30$ W/(m·K).



There is no heat generation within wall B. The left surface of wall B is in contact with wall A. There is no contact thermal resistance at their boundary. The right surface of wall B is exposed to a coolant at $T_\infty = 20$ °C and a convective heat transfer coefficient of 500 W/(m²·K). We insert two temperature sensors to measure the surface temperatures of wall A. The sensor at the interface between the thermal insulation and wall A shows $T_w = 115$ °C. The other sensor at the interface between wall A and wall B shows $T_i = 100$ °C.

(a) Calculate the right surface temperature T_s of wall B.

(b) Calculate the uniform volumetric heat generation $\dot{q}$ at steady state.

(c) Calculate the thermal conductivity k_A of wall A.

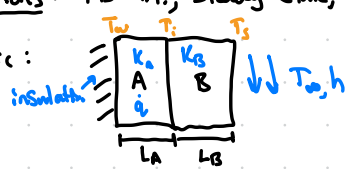
(d) Plot the temperature profile across wall A and wall B and show its important features.

Known: Wall thicknesses L_A, L_B, heat generation $\dot{q}$, thermal conductivities K_A, K_B, Ambient conditions (T_∞, h_{amb}), Wall temperature, interface temperature

Find: T_s (a), $\dot{q}$ (b), K_a (c), plot temperature profile (c)

Assumptions: 1D h.t., Steady state, perfectly insulated wall, const. air and material properties

Schematic:



Analysis:

a) $\dot{q}_{conv} = \dot{q}_{cond,B}$ ← Steady state assumption

$$h\cancel{A}(T_s - T_\infty) = -K_B\cancel{A}\left(\frac{T_s - T_i}{L_B}\right)$$

$$500\frac{W}{m^2\cdot K}(T_s - 20°C) = -\left(30\frac{W}{m\cdot K}\right)\left(\frac{T_s - 100°C}{.1m}\right)$$

$$\boxed{T_s = 50°C}$$

b) $\dot{q}AL_A = \dot{q}_{conv}$ ← Steady state assumption

$$L_A\dot{q}\cancel{A} = h\cancel{A}(T_s - T_\infty)$$

$$\dot{q} = \frac{\left(500\frac{W}{m^2\cdot K}\right)(50° - 20°)}{.3m} = \boxed{50\frac{KW}{m^3}}$$

c) $K_A \frac{\partial^2 T}{\partial x^2} + \dot{q} = 0$ ← Using heat eqn to solve for temperature profile

$$\frac{\partial^2 T}{\partial x^2} = -\frac{\dot{q}}{K_A}$$

$$T(x,\infty) = -\frac{\dot{q}}{2K_A}x^2 + Cx + D$$ $\left.\frac{\partial T}{\partial x}\right|_{x=0} = 0$

$$0 = C$$

$$T(x,\infty) = -\frac{\dot{q}}{2K_A}x^2 + D$$ $T(0,\infty) = 115$

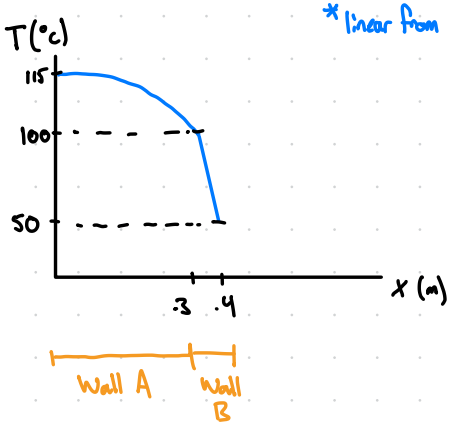
$$115 = D$$

$$T(x,\infty) = -\frac{\dot{q}}{2K_A}x^2 + 115$$

$T(L_A,\infty) = 100$

$$100 = -\frac{50{,}000}{2(K_A)}(.3m)^2 + 115$$

$$\boxed{K_A = 150\ \frac{W}{m \cdot K}}$$

D) $T(x,\infty) = -\frac{\dot{q}}{2K_A}x^2 + 115$ ← In wall A

*linear from $T_1 \to T_s$ in wall B